

Nathaniel Wheeler

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Education

University of Connecticut, Storrs, CT

Bachelor of Science in Electrical Engineering, 08/2023 – 05/2027, **Minor** in Mathematics

Cumulative GPA: 3.90/4.00 | Dean's List [All semesters]

UConn Academic Excellence Scholarships

High School Valedictorian

Skills

Technical Skills: PCB Design, Cadence, SystemVerilog, LTSpice, KiCad, Linux, VHDL, MATLAB, Python, UART, C++

Lab Skills: Oscilloscope, Soldering, Signal Generator, Digital Multimeter, Power Supplies

Work Experience

IBM

Hardware Development Intern (Logic Design), Poughkeepsie, NY

05/26 – 08/26

- Designed custom PCB schematics and digital logic for PCIe hardware validation test vehicles used in AI hardware drawer.
- Collaborated extensively with physical design and signal integrity engineers to ensure proper routing and placement.
- Designed digital logic architecture and developed synthesizable VHDL RTL on a Linux-based environment for a tamper-monitoring circuit used in enterprise server hardware.
- Developed python script to enable large scale read and write operations in I2C using SMBus.

University of Connecticut

08/25 – 06/26

Undergraduate Research Assistant, Storrs, CT

- Designed, simulated, and implemented a battery management control system integrating low-power microcontrollers, NI-DAQ hardware, and analog integrated circuits for battery monitoring and control.
- Performed battery characterization testing to determine cell impedance, state of health, and state of charge of battery.
- Integrated hardware, firmware, and data-acquisition workflows, collaborating across multiple teams to validate system-level electrical performance.

University of Connecticut

08/25 – 06/26

Undergraduate Research Assistant – Bidirectional EV Charger (IFEC), Storrs, CT

- Designed and evaluated rectifier and inverter circuits to maximize efficiency in high power bidirectional EV charger
- Developed logic and physical design for inverter boards
- Tested functionality of converter boards to optimize power and performance requirements, documenting issues and failures in the process.
- Developing and running Simulink models to test and validate control strategies for converter operation.
- Implementing control schemes in both hardware and software testing (MATLAB, ICs, TI CCS)

Raytheon: An RTX Business

Electrical Engineering Intern, Portsmouth, RI

05/25 – 8/25

- Designed electrical schematics and layouts for custom PCBs using KiCad to improve electrical isolation and test performance. Soldered and tested custom PCBs, as well as designed and manufactured enclosures for said boards
- Simulated power amplifier system using MATLAB to assess frequency response and modify input signal based on transfer function
- Designed a control system in Arduino IDE and a custom prototype circuit for a pneumatic fluid dispenser and motor system, improving test results and repeatability
- Assembled and tested multi-element sonar array designed to simulate system and assess issues of the system

Projects

Plant Watering Robot

Spring 2026

- Designed and integrated an autonomous plant watering robot, combining mechanical, electrical, and control systems.
- Modeled power distributions of system in LTSpice, validating voltage regulation and power delivery for all components.
- Designed a custom 4-layer PCB in KiCad for power regulation, MOSFET-based pump driver, and microcontroller interfaces.
- Tested and debugged hardware and firmware to achieve reliable autonomous watering operation.